Scenario-Based Exam Cheat Sheet

IERG4320/IEMS5726 Data Science in Practice | Answer Template + All Scenarios

Answer Template (3 Paragraphs)

情景题万能答题模板，按这三段写

P1: Objectives & Motivations (3%) – What do we want to build? Why does it matter? What is the goal?
"We want to build a [system] to [do what] by using [method]. This is important because [reason 1], [reason 2], and [reason 3]."

P2: Challenges & Limitations (2%) – What makes this hard? What are the weak points?
"The main challenges are [C1], [C2], and [C3]. The limitations are [L1] and [L2]."

P3: Data Collection & Format (3%) – Where do we get data? What does it look like?
"We collect data from [source]. The data is [type] with features like [F1, F2, ...]. The size is about [N] samples with [M] features."

Common Building Blocks

通用答题素材：挑着用

Common Challenges (pick 2–3)

Data quality (missing/noise/outliers) 数据质量 | Class imbalance (use F1/MCC) 类别不平衡 | Cold start 冷启动 | Overfitting (dropout, CV) 过拟合 | High dimensionality (PCA) 高维 | Real-time requirement 实时 | Computational cost (GPU) 计算成本 | Privacy/ethics 隐私伦理 | Domain knowledge 领域知识 | Black swan 黑天鹅

Data Sources & Formats

Sources: API (yfinance, Google Cloud) | DB (SQL, MongoDB) | Web scraping | Public (Kaggle, IMDB, ImageNet) | Sensors (CCTV, mic, IoT) | User logs (clicks, ratings).
Formats: Tabular (CSV, rows×cols) | Time series (OHLCV) | Text (tokens, TF-IDF, embeddings) | Image (H×W×C) | Audio (waveform 1D, spectrogram 2D) | Graph (adjacency + features).

Metrics

Classify: Acc, Prec, Recall, F1, MCC, AUC | Regress: R^2, MAE, RMSE | Cluster: Silhouette↑, CH↑, DB↓ | NLP: BLEU, METEOR, CIDEr, SPICE | ImgGen: FID↓ | Speech: MOS↑ | 3D: Chamfer↓, F-Score↑ | Finance: Sharpe↑, Drawdown↓ | RL: win rate

Scenario Quick Cards

每个场景的要点速查：O=目标 C=挑战 D=数据

A. Prediction & Forecasting

A1. Lottery / Mark Six 彩票预测
O: Use past draw data to predict future numbers with probability and regression models.
C: Totally random; no real pattern; easy to overfit noise; huge number of combinations (C_{49}^{6}=13.98M).
D: Past draw results from official site. Table: date + 6 numbers. Time series format.

A2. Stock Price Prediction 股票预测
O: Use LSTM on past stock data (OHLCV) to predict future price or trend.
C: Sudden crashes (black swan), human emotions affect market, data keeps changing, easy to overfit.
D: yfinance API. Time series: Open, High, Low, Close, Volume. Use 30-day sliding window as input.

A3. Auto Trading Agent 自动交易
O: Train a DQN agent to learn when to buy, sell, or hold to make profit.
C: Reward is sparse (only see profit/loss later), market changes over time, trading fees.
D: yfinance API. State = prices + what we own. Action = buy/sell/hold. Reward = profit or loss.

B. Recommender Systems

B1. Movie/Music Recommendation 电影/音乐推荐
O: Suggest movies or songs that a user will like (Netflix/Spotify style).
C: Cold start: new users have no history; data is very sparse; users may only like one type.
D: User-item rating table (mostly empty). Also: user profiles, item info (genre, tags), click/play logs.

B2. E-commerce Recommendation 电商推荐
O: Suggest products to help users find what they want (Amazon style).
C: Same as B1, plus products change fast, users browse quickly.
D: Purchase history, browsing logs, product details. Mix of table data + text (reviews).

C. NLP Scenarios

C1. Fake News Detection 假新闻检测
O: Tell if a news article is real or fake using NLP.
C: Same word can mean different things; language keeps changing; people try to trick the system.
D: News articles with real/fake labels. Text data. Classic: TF-IDF (25k×5k). DL: BERT tokenizer (max 256 tokens).

C2. Sentiment Analysis 情感分析
O: Tell if a movie review is positive or negative (IMDB).
C: Sarcasm is hard to detect; "not good" vs "good"; word order matters.
D: IMDB: 25k train + 25k test. Text → tokens → embeddings. Two labels: pos/neg.

C3. Chatbot / Assistant 聊天机器人
O: Build an AI that can talk to customers and answer questions.
C: Must understand what user wants; remember past messages; may give wrong answers (hallucination).
D: Chat logs, FAQ pairs (question + answer). Fine-tune GPT or LLaMA.

C4. Translation / Summary 翻译/摘要
O: Translate text between languages, or make a short summary.
C: Different languages have different word orders; idioms hard to translate; output must make sense.
D: Paired sentences (source + target) or documents. Text → tokens. Markov chain as simple baseline.

D. Audio / Speech

D1. Vocal Separation (Karaoke) 人声分离
O: Split a song into voice and music for karaoke.
C: Voice and instruments share same frequency range; hard to separate cleanly.
D: Audio files (WAV/MP3). Sound wave (1D) or spectrogram (2D). Use librosa or Demucs model.

D2. Text-to-Speech (TTS) 语音合成
O: Make computer read text out loud like a real person.
C: Must sound natural, not robotic; show emotion; keep speaker identity.
D: Text + matching audio recordings. Pipeline: text → mel-spectrogram → sound wave. Use Tacotron2 + WaveNet.

Eval: MOS score.

D3. Speech-to-Text (STT) 语音识别

O: Turn spoken words into text (for smart home, voice assistant).
C: Different accents, noisy rooms, people talk differently, words that sound the same.
D: Audio recordings + written transcripts. Sound wave → features. Use HuBERT or Wav2Vec 2.0.

E. Computer Vision

E1. Accident Detection (CCTV) 事故检测

O: Watch CCTV to detect falls, choking in elderly homes or public places.
C: Must work in real-time; things can block the view; lighting changes; false alarms; privacy issues.
D: CCTV video frames. Images (H×W×3). Use CNN (ResNet) to classify. Labels: normal or accident.

E2. Medical Image Diagnosis 医疗影像

O: Find diseases (broken bones, tumors) in X-ray or CT images.
C: Very few sick cases (imbalanced); need doctor labels; must be explainable; small dataset.
D: Hospital image database. Medical images → pixel arrays. Doctor labels. Use pre-trained CNN + fine-tune.

E3. Traffic Surveillance 交通监控

O: Detect cars, accidents, and violations from traffic cameras.
C: Must be real-time; bad weather/light; cars block each other; many vehicle types.
D: Traffic camera video. Video frames. Use object detection (Mask R-CNN). Labels: car type, violation.

E4. Image Captioning 图像描述

O: Write a sentence to describe what is in a picture (help blind people).
C: Must find objects + actions + how they relate + write correct grammar.
D: Image + caption pairs (COCO dataset). Image: 224×224×3. Use CNN+RNN or ViT+Transformer. Eval: BLEU, CIDEr.

F. Generation

F1. AI Image Generation AI图片生成

O: Create pictures from text descriptions for art or ads.
C: Output is very high-dim (millions of pixels); must fill in background; copyright issues.
D: Text-image pairs. Type text prompt → use DALL-E or Stable Diffusion. Eval: FID. CLIP scores the results.

F2. DeepFake Detection 深度伪造检测

O: Find fake videos to stop scams and misinformation.
C: Fakes look more and more real; best AI only catches 80%; attackers keep improving.
D: Real + fake video pairs. Face crops from video frames. Binary: real or fake. Look at: blinking, lip sync, skin.

F3. 2D to 3D Model 2D转3D

O: Turn a single photo into a 3D model (for games, architecture).
C: Many possible 3D shapes match one 2D view; needs learned priors; may have floating artifacts.
D: RGB images + 3D ground truth. Use TripoSR: image → encoder → transformer → NeRF → mesh. Eval: Chamfer Distance, F-Score.

F4. Text to 3D Model 文字转3D

O: Type words and get a 3D model (Shap-E by OpenAI).
C: Less detailed than hand-made models; shape errors; floating points; limited by 2D image quality.
D: Text + 3D pairs. Prompt → image diffusion → point cloud diffusion → mesh. About 13 seconds each.

G. Game & RL

G1. Card Game AI (Blackjack) 21点AI

O: Train an AI to play Blackjack well using RL.
C: Only get reward at game end (sparse); cards are random; must balance try-new vs use-best.
D: Simulated games. State = (my total, dealer card, usable ace). Actions: hit or stand. Use D3QN, A3C, or AlphaZero.

G2. Board Game AI 棋类AI

O: Train AI to master Chess or Go by playing against itself (AlphaZero).
C: Huge number of possible positions; need to plan many moves ahead; MCTS is expensive.
D: Self-play generated games. State = board position. Use MCTS + dual net (policy head + value head).

G3. Multi-Game Agent 多游戏智能体

O: One AI that can play many different games without forgetting old ones.
C: Catastrophic forgetting: learning a new game erases what it learned before.
D: Multiple game environments. Use EWC to protect important weights. Train: game A → game B (with EWC) → game C.

H. Robot Learning

H1. Robot Walking 机器人行走

O: Teach a robot to walk like a human using RL + transfer learning.
C: Many joints to control at once; must keep balance; simulation differs from real world.
D: Simulated physics environment. State = joint angles + body pose. Learn step by step: crawl → walk with wall → walk freely.

H2. Robot Grasping 机器人抓取

O: Teach a robot arm to pick up objects.
C: Objects come in all shapes; needs high precision; soft objects are tricky.
D: Camera images + robot arm joint data. Use imitation learning, inverse RL, or diffusion policies.

Which Model for Which Problem?

问题类型→模型速查

Problem	Models
Classification	DT, RF, SVM, KNN, ANN/MLP
Regression	Linear, Ridge, LASSO, SVR, ANN
Clustering	K-Means, Agglomerative, Spectral
Recommendation	CF, Content, Hybrid, MF/SVD, Wide&Deep, GCN
Time series	EMA, LSTM, RNN
Text classify	TF-IDF+NB, NBoW, BERT, TinyBERT
Text generation	Markov, GPT, LLaMA
Speech (TTS)	WaveNet, Tacotron2
Speech (STT)	Wav2Vec 2.0, HuBERT
Image classify	AlexNet, VGG, ResNet, ViT
Image caption	CNN+RNN+Attn, ViT+Transformer
Image generation	GAN, DALL-E, Diffusion
3D generation	TripoSR, Shap-E
Game agent	Q-Learning, DQN, D3QN, A3C, AlphaZero
Robot control	RL, imitation, behavior cloning, inverse RL